# Perceived Stress Tracks the Tinnitus Symptom More Than the Audiometric Threshold: A Survey-Weighted KNHANES Study (STARS)

## With a Prespecified Cross-National Validation Plan and a Clinician-Governed Referral-Safety Component

Geunsik Lim[1]

[1]Sungkyunkwan University, Republic of Korea; leemgs@g.skku.edu

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| **Article type:** | Research Article (survey-weighted public-data analysis) with a prespecified analysis plan |
| **Target journal:** | *American Journal of Audiology* (AJA) |
| **Running head:** | Perceived Stress, Tinnitus, and Audiometric Thresholds |
| **Reporting guidelines:** | STROBE (primary analysis); TRIPOD+AI, SPIRIT (prospective components) |
| **Primary findings:** | Real KNHANES perceived-stress results (Table 1); supporting NHANES results (Table 3) |
| **Prospective components:** | Cross-national validation, clinician-governed AI extraction, and SSNHL referral-safety layer (plan only; no results) |
| **Abstract:** | Structured, ∼250 words |
| **Body word count:** | ∼3,800 (excludes abstract, references, tables; prospective protocols and methods-detail tables in Supplementary Materials) |
| **Tables:** | 3 main + 3 supplementary **Figures:** 2 **Supplement:** `supplement.tex` (Supplementary Methods S1–S2, Tables S1–S3) |
| **Open repository:** | https://github.com/leemgs/stars-audiology |

*Note.* The primary KNHANES perceived-stress analysis is reported as a complete empirical result; the cross-national validation, AI-extraction, and clinical-extension components remain prospective (design only).


## Abstract

**Purpose:** STARS tests, in nationally representative survey data, whether *perceived stress* dissociates the tinnitus *symptom* from the audiometric *threshold*. Analyses, variables, cycles, and adjustment sets were prespecified. The exposure is perceived stress (a single general item), not occupational stress, and every estimate is an association, not a cause.

**Method:** KNHANES V (2010–2012; adults 40–69) is the primary cohort because it carries a general perceived-stress item; U.S. NHANES (2011–2018) provides supporting analyses under a PHQ-9 distress proxy. A directed acyclic graph fixes the minimal-sufficient adjustment set (age, sex, occupational noise) and separates confounders from mediators. Outcomes are modeled with survey-weighted (Taylor-linearized) regression, per STROBE.

**Results (KNHANES):** High perceived stress was associated with *tinnitus* (OR 1.42, 1.26–1.60) but only weakly and non-significantly with the *audiometric threshold* (OR 1.18, 0.99–1.41), which was dominated by noise and age—the prespecified dissociation (H1). The tinnitus association survived adjustment for depressed mood and audiometric hearing loss and persisted for non-bothersome tinnitus. Excluding conductive loss by tympanometry, the threshold association strengthened to significance (OR 1.29, 1.07–1.55), so the dissociation is one of degree. NHANES showed the same directional pattern (directional consistency, not validation).

**Conclusions:** In representative Korean adults, perceived stress tracks the tinnitus symptom more than the audiometric threshold once noise and age are accounted for—consistent with a partly central, non-cochlear correlate of tinnitus. Findings are reproducible from committed code; because the design is cross-sectional, directionality (including reverse causation) is unresolved, and prospective cohorts remain necessary.

**Keywords:** tinnitus; bothersome tinnitus; hearing loss; audiometry; perceived stress; work-related factors; KNHANES; NHANES; survey-weighted analysis; reproducibility; audiology.

# 1 Introduction

Tinnitus and hearing loss are among the most common auditory health concerns worldwide, affecting communication, sleep, mood, work participation, and quality of life. More than 1.5 billion people live with some hearing loss, a number projected to rise by 2050 (World Health Organization, 2021), and tinnitus affects an estimated 10–15% of adults, a meaningful subset of whom experience bothersome tinnitus (Tunkel et al., 2014; McCormack et al., 2016). In employed middle-aged adults, tinnitus frequently emerges alongside psychosocial stress, sleep disruption, noise, and delayed help-seeking; because these factors are intertwined, simple causal narratives are unreliable and can mislead patients and clinicians.

## 1.1 Motivating observation and naming principle

Consider an anonymized composite trajectory: a worker in their late 40s develops unilateral tinnitus during heavy workload, which resolves over months; about a year later the other ear develops sudden fullness and reduced audibility, but— unaware of the narrow treatment window for sudden sensorineural hearing loss (SSNHL)—the person delays care, and recovery is incomplete. Two clinically actionable points follow: stress-associated tinnitus is common and often self-limiting, but sudden or rapidly progressive hearing loss is an urgency in which delayed care, not stress, drives irreversible impairment. Attributing acute hearing symptoms to “stress” without assessment is hazardous. Accordingly, STARS (Stress, Tinnitus, and Audiometric Research Study) adopts a strict naming principle: because KNHANES and NHANES carry general perceived-stress and mood items rather than validated job-strain instruments, we never call the exposure measured “occupational stress.” It is labeled *perceived stress*, with *work-related factors* (occupation, employment, and where available hours and shift work) modeled as distinct covariates.

## 1.2 A survey-weighted study reported against a prespecified plan

This is a survey-weighted study whose *primary* hypothesis test—the perceived-stress analysis in KNHANES—is reported in full (Section 5), with supporting NHANES analyses. To limit analyst degrees of freedom, the design, variable definitions, cycles, adjustment sets, and evaluation plan were fixed *before* analysis, and committed code and outputs make every value reproducible. Three further components—cross-national validation, clinician-verified AI extraction, and a clinical extension—are *prospective* (design only) and kept subordinate to the empirical finding.

## 1.3 Aims and hypotheses

Aim 1 is the empirical contribution reported here; Aim 3 is developed *with results* in the companion SAFE-EAR paper, Aim 4's open pipeline accompanies this article, and Aim 2 is prospective.

1. **Association (development).** Estimate weighted associations between perceived stress and (a) tinnitus and (b) audiometric hearing loss in KNHANES, adjusting for noise and health covariates. *H1: a positive stress–tinnitus association persists after adjustment, while the association with audiometric thresholds is weaker and may attenuate to non-significance after adjusting for noise and age.*

2. **Transportability.** Test whether harmonized models developed in KNHANES retain discrimination and calibration in NHANES. *H2: discrimination degrades modestly but calibration is restorable by recalibration.*

3. **Explainable, safe AI (companion paper).** Evaluate whether schema-constrained extraction feeding a deterministic red-flag layer routes SSNHL and neurologic warning signs to urgent referral. *H3: near-complete red-flag recall, bounded by extraction quality.* Developed with results in SAFE-EAR.

4. **Open, shared benefit.** Provide an openly licensed, reproducible pipeline (code, data dictionary, model cards, prompts).

## 1.4 Contributions

Primary: a survey-weighted, public-data result showing that perceived stress dissociates the tinnitus *symptom* from the audiometric *threshold*, reported without unsupported causal claims and reproducible end-to-end. Secondary: an open, reproducible scaffold (harmonized mappings, design-based estimators, model cards, safety tests) that lets others regenerate every number and add local cohorts. The remaining components—a prespecified KNHANES→NHANES external-validation plan and a clinically governed AI/referral-safety component in which a deterministic red-flag layer overrides probabilistic predictions—are prospective (design only); any prediction model is a research risk-stratification tool, never a screening, diagnostic, or triage device.

# 2 Related Work

## 2.1 Epidemiology and prior KNHANES work

Tinnitus and hearing loss often coexist but relate heterogeneously. KNHANES analyses have estimated Korean tinnitus prevalence and correlates (Park et al., 2014; Joo et al., 2015), and NHANES has enabled U.S. studies of audiometry, occupational noise, tinnitus, and mental-health correlates (Hoffman et al., 2017; Chakrabarty et al., 2024; Mahboubi et al., 2013); reviews emphasize heterogeneous case definitions and the need for harmonized variables when pooling across surveys (McCormack et al., 2016). Evidence linking psychological stress to tinnitus is consistent but largely observational, with residual confounding by noise, sleep, and cardiometabolic factors difficult to exclude (Baigi et al., 2011). On the same KNHANES cycles, prior work has examined tinnitus alongside distress—most closely Lee et al. (2018), on higher perceived stress and mood symptoms in premenopausal women with tinnitus. STARS is therefore *not* the first to link tinnitus with distress in KNHANES; its contribution is methodological: (i) a *prespecified, falsifiable* contrast between the tinnitus *symptom* and the audiometric *threshold* (H1) rather than a single tinnitus–distress association; (ii) an explicit DAG that separates confounders from mediators, reporting both minimal-sufficient and mediator-adjusted models; (iii) a cross-national directional-consistency check in NHANES under a related distress construct; and (iv) a fully reproducible open scaffold. The novelty is design discipline and reproducibility, not the mere existence of a tinnitus–stress association.

## 2.2 Beyond the pure-tone average

Because real-world listening difficulty and tinnitus burden can persist when the better-ear pure-tone average appears near-normal, STARS separates tinnitus presence from bothersome tinnitus and analyzes worse-ear and high-frequency thresholds and asymmetry, not better-ear PTA alone.

## 2.3 Clinical pathways and reporting standards

Practice guidelines frame the endpoints. The tinnitus guideline distinguishes bothersome from non-bothersome tinnitus and prioritizes education, cognitive-behavioral strategies, and sound therapy (Tunkel et al., 2014), while the sudden-hearing-loss guideline defines a "golden window"—prompt audiometric confirmation and early corticosteroid therapy within roughly two weeks—so delayed presentation is associated with worse outcomes (Chandrasekhar et al., 2019). This is the clinical rationale for the deterministic red-flag referral-safety layer developed in the companion SAFE-EAR paper. The observational analyses adhere to STROBE (von Elm et al., 2007); the prospective prediction and open-medical-LLM components follow the TRIPOD+AI statement (Collins et al., 2024).

# 3 Public Dataset Selection

Dataset selection was governed by four requirements for the primary association analyses: (i) survey-weighted population representativeness; (ii) objective pure-tone audiometry; (iii) tinnitus and/or hearing-difficulty items; and (iv) covariates for noise exposure, psychosocial stress, and general health. Only KNHANES and NHANES jointly satisfy all four at population scale; other public resources are retained as optional secondary datasets. Supplementary Table S1 summarizes the selection, and Supplementary Table S2 maps target variables to each source.

## 3.1 Primary Development Dataset: KNHANES

KNHANES is selected as the primary development dataset because it is nationally representative for Korea and has previously supported tinnitus and hearing-loss analyses (Park et al., 2014; Joo et al., 2015). Its advantages include Korean population relevance, complex-survey weights, pure-tone audiometry in applicable cycles (otologic examination cycles 2010–2012), tinnitus and hearing-difficulty questionnaire items in selected cycles, health and behavioral covariates, and psychosocial or stress-related self-report variables (e.g., perceived-stress and depressed-mood items). It also provides a realistic bridge to future Korean hospital cohorts, including the planned prospective clinical extension. Because item wording and audiometry availability differ across cycles, the analysis prespecifies a per-cycle codebook review and restricts pooling to cycles with comparable definitions.

The limitations are equally important. KNHANES is not a workplace cohort, and the audiometric analyses are cross-sectional. It can therefore evaluate *associations* among perceived stress, work-related variables, tinnitus, and audiometric outcomes, but cannot by itself establish that occupational stress *caused* tinnitus or hearing loss. Treatment timing and recovery after sudden hearing loss are not captured.

## 3.2 External-Validation Dataset: NHANES

NHANES is selected as the primary external-validation dataset because it provides openly downloadable U.S. public-use files, pure-tone audiometry and tympanometry in selected cycles (e.g., 2011–2012, 2015–2016, 2017–March 2020), occupational and recreational noise-exposure items, tinnitus and hearing questionnaires in selected cycles, and extensive demographic and health covariates. NHANES enables testing whether feature definitions and risk models derived from KNHANES retain discrimination and calibration across population, language, health-system, and occupational contexts. As with KNHANES, complex-survey weights, strata, and primary sampling units must be honored, and cycle-specific age eligibility for the audiometry module must be respected.

### 3.3 Prospective Clinical Extension (Hospital-Based Cohort)

A prospective otolaryngology/audiology cohort is specified to address what public cross-sectional data cannot—treatment timing, SSNHL recovery, longitudinal tinnitus burden, and rehabilitation—and to supply deidentified audiograms and clinical narratives for the companion clinical validation and safety evaluations (Supplementary Methods S2). Raw clinical data are not redistributed; only derived, deidentified, or synthetic artifacts are shared, subject to IRB approval.

## 4 Methods (Prespecified)

The primary observational analysis reported here follows STROBE (von Elm et al., 2007), with its completed checklist accompanying the manuscript; the prospective prediction and clinical components follow TRIPOD+AI (Collins et al., 2024) and SPIRIT, respectively, and are developed in the companion SAFE-EAR paper. To limit analyst degrees of freedom, all design choices below were fixed prior to real-data analysis.

### 4.1 Study Design

The work has three stages, summarized in the conceptual framework of Figure 1. **Stage 1** analyzes KNHANES as the primary public-data development cohort (association estimation and research-model development). **Stage 2** evaluates cross-national transportability in NHANES (geographic external validation). **Stage 3** prepares a prospective hospital-based clinical extension for clinical external validation, treatment-timing analysis in SSNHL, and evaluation of the AI extraction and safety components. Stages 1–2 use only deidentified public-use files; Stage 3 proceeds only under IRB approval.

## 4.2 Fixed Datasets, Cycles, and Eligibility

To eliminate analyst degrees of freedom, the survey cycles, files, ages, frequencies, and exclusions are fixed in advance (Supplementary Table S1). **KNHANES:** the otologic-examination cycles carrying pure-tone audiometry and tinnitus items (KNHANES V, 2010–2012) form the primary development window; cycles are pooled only after confirming identical item wording and audiometry protocols. This window is dictated by data availability: the KNHANES ear/hearing examination was fielded only in 2008–2012 and discontinued thereafter, so 2010–2012 are the most recent Korean cycles pairing the perceived-stress item with audiometry (a data-vintage caveat noted in the Discussion). **NHANES:** the audiometry-module cycles 2011–2012, 2015–2016, and 2017–2018 form the external-validation window. For both, eligible records are adults aged 40–69 years (the band jointly covered by the audiometry modules and our target of mid-to-late-career workers), with valid examination weights and a non-missing value for the outcome under analysis. Pure-tone thresholds are analyzed at 0.5, 1, 2, 3, 4, 6, and 8 kHz per ear; conductive patterns are flagged and, where tympanometry/air–bone data permit, excluded in a sensitivity analysis. We prespecify expected raw and analytic sample sizes per window (reported in the repository data dictionary) and the exact set of predictors available in *both* surveys; any predictor available in only one survey is excluded from the common transportable model and used only in country-specific extended models.

## 4.3 Exposure: Perceived Stress and Work-Related Factors

Consistent with the naming principle (Introduction), the primary exposure is *perceived stress*, operationalized from each survey's general stress/psychological-distress item, and *work-related factors* (occupation category, employment status, and—where a cycle provides them—working hours and shift work) are modeled as separate covariates or effect modifiers, never as a proxy for validated job strain. Because item wording, response categories, and cultural response styles differ across surveys, Supplementary Table S3 records, for every exposure and covariate, the original question, response scale, harmonization rule, and a measurement-comparability rating (high/medium/low). Variables rated *low* comparability (typically the stress and some psychosocial items) are (a) excluded from the common transportable model, (b) entered only in country-specific models, and (c) examined in a formal measurement-invariance check (configural/metric where a latent construct is defined). We do not pool a variable across countries unless at least metric-level comparability is plausible.

## 4.4 Causal Structure and Adjustment Sets

Figure 2 presents the directed acyclic graph (DAG) encoding our assumptions. Age, sex, education, occupation/employment, and noise exposure are treated as confounders of the perceived-stress→outcome relationships and constitute the **minimal sufficient adjustment set** for the primary analysis. Depressed mood and sleep are treated as *potential mediators* of the stress→tinnitus path (and possibly colliders when conditioning opens non-causal paths); conditioning on them can constitute overadjustment. Accordingly:

- **Primary model** adjusts for the minimal sufficient set only (age, sex, education, occupation/employment, noise).
- **Extended model** additionally includes depressed mood, reported separately and interpreted as a mediator-adjusted (not confounder-adjusted) estimate; a **fully adjusted** sensitivity model further adds socioeconomic status (education, income) and, where available, smoking and cardiometabolic comorbidity, to address residual confounding by conventional covariates.
- **Mediation** is summarized by the mediator-adjusted estimate together with an exploratory difference-method proportion (the attenuation of the perceived-stress log-odds after adjusting for depressed mood); we do not claim a formal counterfactual (direct/indirect) decomposition, because cross-sectional data and logistic non-collapsibility make such estimates unreliable.

- **Collider check** enumerates conditioning sets that could induce collider bias and excludes them from the primary specification.

Because restricting to employed participants can induce a healthy-worker effect (and a selection collider), the primary population is *all eligible adults*, with an *employed-restricted* analysis reported as a prespecified secondary contrast rather than the main result.

## 4.5 Outcomes

Endpoints are defined a priori and separated:

- **Tinnitus presence** (any tinnitus in the past 12 months) and **bothersome tinnitus** (tinnitus causing annoyance/sleep or concentration difficulty) are modeled as *distinct* endpoints, since their determinants differ.
- **Hearing loss** is quantified by the speech-frequency pure-tone average ($PTA_{0.5,1,2,4}$) for *both the better ear and the worse ear*, plus a **per-ear** (ear-level, mixed-effects) analysis, because a better-ear summary can mask the unilateral/asymmetric presentations most relevant to SSNHL. The primary threshold is PTA $>$ 25 dB HL (the WHO grade-1 cutoff (World Health Organization, 2021)); $>$ 20 dB HL and high-frequency $PTA_{3,4,6}$ are prespecified sensitivity definitions, with the 25 dB HL choice justified as the established clinical cutpoint.
- **Asymmetric hearing loss** is defined as an interaural difference $\geq$15 dB at $\geq$2 adjacent frequencies, consistent with asymmetry criteria used in sudden/asymmetric hearing-loss guidance (Chandrasekhar et al., 2019).
- **Secondary:** self-reported hearing difficulty, hearing-aid use, and quality-of-life correlates.

**Noise exposure** is coded, where cycles permit, not merely as a binary but with duration, intensity/source (occupational, recreational, firearm), and hearing-protection use, so residual confounding by noise is minimized.

## 4.6 Statistical Analysis

All public-data analyses use survey-aware methods (Taylor-linearization variance estimation honoring weights, strata, and PSUs). The analytic age band (40–69) is estimated as a *subpopulation (domain)* that retains the full survey design rather than by subsetting the data before variance estimation, and single-PSU ("lonely") strata are handled by the standard conservative centering option (centering the lone unit at the grand mean) rather than by dropping the stratum term, which would under-estimate variance. Estimates are reproducible from the committed, unit-tested Python estimators and, independently, from the R `survey` package (`code/src/reproduce_survey.R`); domain estimation and lonely-PSU centering left the headline confidence intervals essentially unchanged (e.g., the tinnitus OR remained 1.42, 95% CI 1.26–1.60). Multiplicity across the prespecified secondary contrasts is controlled by Benjamini–Hochberg false-discovery-rate–adjusted $q$-values (reported with the sensitivity models), with the two primary endpoints reserved as the confirmatory H1 test. Descriptive analyses estimate weighted prevalence with 95% confidence intervals. Association analyses use weighted logistic, ordinal, or linear regression by outcome type. The **primary specification adjusts for the minimal sufficient set**; the extended (mediator-adjusted) model is secondary. Results are reported as associations (odds ratios or mean differences) with 95% CIs, explicitly not as causal effects.

**Missing data.** Missingness is described by variable and mechanism. The KNHANES tinnitus/audiometry outcomes come from the otology (ENT) examination, a $\geq$40 sub-sample; because the KDCA does not release an ENT-examination-specific weight for these cycles, we use the distributed interview-plus-examination weight (`wt_itvex`), pooled across 2010–2012 by dividing by the number of years per KDCA guidance, and treat any residual calibration mismatch for the ENT sub-sample as a weighting limitation (Discussion). The primary analysis is *complete-case* (design-based estimation on records with the outcome and model covariates observed); because item missingness in the analytic variables is low and the design-based estimators handle it directly, complete-case is the primary estimator and multiple imputation by chained equations (MICE, missing-at-random, Rubin's-rules pooling) is a prespecified robustness check. In external validation, records missing a common predictor follow the same rule in both surveys.

**Sample size and precision.** Sample sizes are fixed, so we frame precision rather than post-hoc power: with several thousand audiometry participants per window, weighted prevalence is estimable to within a few percentage points and moderate associations (OR $\approx$ 1.3–1.5) are detectable at conventional error rates. Exact expected analytic $N$ per window is fixed in the repository before analysis.

## 4.7 Prospective prediction and safety components (companion paper)

Beyond the associations reported here, STARS prespecifies two *research-only* risk-stratification targets (tinnitus, hearing loss) and a deterministic, guideline-derived red-flag referral-safety layer for sudden sensorineural hearing loss. These prospective components—never screening, diagnostic, or triage devices—are developed *with results*, with their intended-use specifications, metrics, and an open safety benchmark, in the companion SAFE-EAR paper and carry no results here.

## 4.8 Prospective external validation (companion program)

Cross-national transportability of a *fitted* KNHANES model to NHANES (Aim 2) is prospective and carries no results here; the NHANES analyses reported below are directional support under a distinct distress proxy, not a validation of a transported model. The prespecified transportability protocol—weighted versus unweighted performance, calibration and recalibration of the frozen model, common-support restriction, case-mix standardization, a reduced-vs-extended model comparison, and prespecified subgroup fairness—is given in

Supplementary Methods S1 and the repository.

### 4.9 Sensitivity Analyses

Prespecified sensitivity analyses test alternative hearing-loss definitions (25 vs. 20 dB HL; high-frequency PTA; worse-ear and per-ear), age restrictions, exclusion of conductive patterns when tympanometry permits, alternative stress operationalizations, mediator- vs. confounder-treatment of depressed mood/sleep, all-adult vs. employed-restricted populations, and stratification by noise exposure, sex, and asymmetry.

### 4.10 Ethics and Governance

Stages 1–2 use publicly available, deidentified survey data and do not constitute human-subjects research requiring new consent; KNHANES and NHANES were approved by their respective review boards, and their use follows each provider's data-use terms. The Stage-3 clinical extension will obtain IRB approval at the participating clinical site, with a data-management plan covering deidentification, access control, and secondary-use consent. No identifiable patient data are placed in the public repository; only derived, deidentified, or synthetic artifacts are shared.

## 5 Results

We report the *primary* perceived-stress analyses in the KNHANES development cohort, then the *supporting* NHANES analyses. KNHANES is the primary test because, unlike NHANES, it carries a general perceived-stress item (the NHANES PHQ-9 proxy is a DAG *mediator*). All estimates are design-based (survey-weighted, Taylor-linearized) associations, not causal effects, and every value below is reproduced by the committed `outputs/knhanes_results.json` and `nhanes_results.json`.

## 5.1 Primary results (KNHANES, adults 40–69)

Pooling the otologic-examination cycles 2010–2012 (KNHANES V; analytic domain $n$ = 10,467; complete-case models retained 9,142–9,532 by outcome), survey-weighted prevalence was 21.1% (95% CI 19.9–22.3) for any tinnitus, 7.0% (6.3–7.7) for bothersome tinnitus, 15.4% (14.5–16.4) for better-ear speech-frequency hearing loss (>25 dB HL), and 24.3% (23.2–25.3) for high perceived stress.

Table 1: **Real KNHANES results** (Korean adults 40–69). Primary development cohort with the perceived-stress exposure. Prevalence is design-based (survey-weighted, Taylor-linearized); associations are mutually adjusted design-based odds ratios and are *not* causal. Generated by `code/src/knhanes_analysis.py`. $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$.

| | Tinnitus | Hearing loss (>25 dB HL) |
|---|---|---|
| Prevalence (95% CI) | 21.1% (19.9–22.3) | 15.4% (14.5–16.4) |
| *Adjusted associations, OR (95% CI)* | | |
| Perceived stress (high) | 1.42 (1.26–1.60)*** | 1.18 (0.99–1.41) |
| Occupational noise | 1.37 (1.15–1.63)*** | 1.72 (1.44–2.06)*** |
| Age (per year) | 1.03 (1.02–1.04)*** | 1.13 (1.12–1.14)*** |
| Female sex | 1.08 (0.96–1.22) | 0.47 (0.40–0.55)*** |

**Primary association (H1).** In the minimal-sufficient model (perceived stress, occupational noise, age, sex), high perceived stress was associated with *tinnitus* (OR 1.42, 1.26–1.60) but only weakly and non-significantly with *better-ear hearing loss* (OR 1.18, 0.99–1.41), where occupational noise (OR 1.72, 1.44–2.06) and age (OR 1.13 per year) dominated. This is the prespecified symptom-versus-threshold dissociation (H1): perceived stress tracks the tinnitus *symptom* ($p < 0.001$) but attenuates against the audiometric *threshold* ($p = 0.07$) once occupational noise and age are accounted for.

**Sensitivity analyses.** The stress–tinnitus association survived adjustment for the mediator depressed mood (OR 1.29, 1.13–1.47) and for audiometric hearing loss itself (OR 1.44, 1.27–1.64), so it is not an artifact of under-adjustment for the strongest tinnitus correlate (Table 2). It persisted for *non-bothersome* tinnitus versus none (OR 1.24, 1.05–1.45) and was strongest for bothersome tinnitus (OR 1.79, 1.46–2.21); across seven secondary contrasts we control the false-discovery rate (Benjamini–Hochberg), and these remain significant ($q < 0.02$). The dissociation is *relative, not absolute*: perceived stress was weakly associated with the *worse-ear* threshold (OR 1.15, 1.00–1.31; does not survive FDR, $q \approx 0.07$) but not with the high-frequency definition (OR 1.07, 0.93–1.23). Because the KNHANES examination provides air-conduction thresholds and tympanometry but not bone conduction, we excluded middle-ear pathology using normal bilateral tympanometry: the stress–threshold association then *strengthened* to significance (OR 1.29, 1.07–1.55) while the tinnitus association was essentially unchanged (OR 1.42, 1.25–1.62), indicating the full-sample null was partly a conductive-contamination artifact and that the contrast is one of *degree*.

Table 2: **Extended and sensitivity models (KNHANES, adults 40–69).** Perceived-stress odds ratio (95% CI) across prespecified specifications, isolating the robustness of the primary exposure. Rows add the mediator (depressed mood) or audiometric hearing loss to the minimal-sufficient set (perceived stress, occupational noise, age, sex), and vary the hearing-loss outcome definition (better-ear, worse-ear, high-frequency). All are design-based associations, not causal effects; mediator-adjusted rows are interpreted as mediator- (not confounder-) adjusted. Bothersome tinnitus (moderate/severe annoyance) is modeled versus all others; non-bothersome tinnitus (present but not annoying) versus no tinnitus, excluding bothersome cases. Secondary contrasts are Benjamini–Hochberg false-discovery-rate controlled (reported in text). Generated by `code/src/knhanes_analysis.py`. $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$.

| Model specification | Perceived stress, OR (95% CI) | $N$ |
|---|---|---|
| Tinnitus — minimal-sufficient (primary) | 1.42 (1.26–1.60)*** | 9504 |
| + depressed mood (mediator-adjusted) | 1.29 (1.13–1.47)*** | 9440 |
| + audiometric hearing loss | 1.44 (1.27–1.64)*** | 9142 |
| + SES/comorbidity (fully adjusted) | 1.29 (1.13–1.48)*** | 9307 |
| Tinnitus, normal tympanometry (conductive-excluded) | 1.42 (1.25–1.62)*** | 8510 |
| Hearing loss, better ear — minimal-sufficient | 1.18 (0.99–1.41) | 9176 |
| + depressed mood (mediator-adjusted) | 1.11 (0.92–1.34) | 9131 |
| Hearing loss, worse ear — minimal-sufficient | 1.15 (1.00–1.31)* | 9176 |
| Hearing loss, high-freq (3/4/6 kHz) — minimal-sufficient | 1.07 (0.93–1.23) | 9176 |
| Hearing loss, normal tympanometry (conductive-excluded) | 1.29 (1.07–1.55)** | 8217 |
| Bothersome tinnitus — minimal-sufficient | 1.79 (1.45–2.21)*** | 9532 |
| Non-bothersome tinnitus vs none (reverse-causation sensitivity) | 1.24 (1.05–1.45)* | 8781 |

The employed-restricted (healthy-worker) contrast, the KNHANES→NHANES external validation (H2), and subgroup fairness are prospective components that follow the prespecified plan (Methods; Supplementary Methods S1) and carry no results here; the open-LLM extraction and referral-safety layer are developed, with results, in the companion SAFE-EAR paper.

### 5.2 Supporting results (NHANES): directional consistency

For U.S. adults aged 40–69 (pooling 2011–2012, 2015–2016, 2017–2018; Table 3), survey-weighted prevalence was 28.5% (26.5–30.4) for any tinnitus, 12.0% (10.4–13.6) for better-ear hearing loss, and 8.4% (7.5–9.4) for a positive PHQ-9 screen. In mutually adjusted design-based models, *tinnitus* was associated with occupational noise (OR 1.59, 1.32–1.92), depressed mood (OR 1.61, 1.23–2.11), and female sex (OR 1.61, 1.35–1.91) but not age; *hearing loss* was dominated by age (OR 1.11 per year, 1.09–1.13) and depressed mood (OR 2.29, 1.56–3.35), inversely by female sex (OR 0.42, 0.29–0.61), and only weakly by noise (OR 1.22, 0.94–1.58). Because the PHQ-9 proxy is a DAG mediator and NHANES lacks the perceived-stress exposure, these mediator-level associations reproduce the symptom-versus-threshold pattern in a different population but do not substitute for the primary KNHANES analysis.

## 6 Discussion

STARS reports one empirical finding—a symptom-versus-threshold dissociation—and builds a prespecified companion program around it.

### 6.1 Interpretation of the primary finding

The claim is *not* that stress damages hearing. Perceived stress is associated with tinnitus (OR 1.42), and this survives adjustment for occupational noise, age, the mediator depressed mood (OR 1.29), and measurable hearing loss itself (OR 1.44, one of the strongest tinnitus correlates), so it is not an artifact of under-adjustment. Its association with the audiometric threshold, by

Table 3: **Real NHANES results** (U.S. adults 40–69; pooled 2011–2012, 2015–2016, 2017–2018). Prevalence is design-based (survey-weighted, Taylor-linearized); associations are odds ratios (OR) from design-weighted logistic regression, mutually adjusted, and are *not* causal. Depressed mood is a PHQ-9 screen used here as the available psychological-distress proxy (NHANES lacks a validated perceived-stress instrument); per the study DAG it is a mediator, so its OR is a mediator-level association. Generated by `code/src/nhanes_analysis.py`. $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$.

| | Tinnitus | Hearing loss (>25 dB HL) |
|---|---|---|
| Prevalence (95% CI) | 28.5% (26.5–30.4) | 12.0% (10.4–13.6) |
| *Adjusted associations, OR (95% CI)* | | |
| Occupational noise | 1.59 (1.32–1.92)*** | 1.22 (0.94–1.58) |
| Depressed mood (PHQ-9≥10) | 1.61 (1.23–2.11)*** | 2.29 (1.56–3.35)*** |
| Age (per year) | 1.00 (0.98–1.01) | 1.11 (1.09–1.13)*** |
| Female sex | 1.61 (1.35–1.91)*** | 0.42 (0.29–0.61)*** |
| Analytic $N$ (associations) | 4603 | 4463 |
| Depression (PHQ-9≥10) prevalence: 8.4% (7.5–9.4), $N = 7293$ | | |

contrast, attenuates to non-significance (OR 1.18, $p = 0.07$) once noise and age are accounted for—best read as a *weak, attenuated* association whose confidence interval barely includes the null, not as "no association." The contrast is therefore one of degree and precision: perceived stress relates to tinnitus perception and burden—a partly central, attention-modulated phenomenon—more than to cochlear thresholds. It is *relative, not absolute*: a worse-ear signal exists (OR 1.15) but does not survive false-discovery-rate adjustment, and excluding conductive loss by tympanometry strengthens the threshold association (OR 1.29). At a stress prevalence near one in four adults, OR 1.42 is a modest but population-relevant association, not a large effect.

## 6.2 Directionality, measurement, and NHANES

The chief threat is *reverse causation*: a cross-sectional odds ratio cannot separate "stress associated with having tinnitus" from "tinnitus generating stress," and bothersome tinnitus is itself a documented stressor. Two prespecified analyses bound—without eliminating—this. The association *persists* for non-bothersome tinnitus versus none (OR 1.24; $q < 0.02$), which is harder to explain by feedback alone; yet the monotone gradient toward bothersome tinnitus (OR 1.79) is consistent with a feedback contribution. Resolving temporality requires the prospective cohort. Both exposure and outcome are *single self-report items* (perceived stress, not a validated PSS-10 or job-strain scale; tinnitus, not THI/TFI); if this misclassification is non-differential—the usual expectation—it biases estimates *toward the null*, so OR 1.42 is a floor, not a ceiling. NHANES provides *directional consistency, not external validation*: it carries no perceived-stress item, its PHQ-9 proxy is a DAG mediator, and it reproduces the symptom-versus-threshold *pattern* in a different population—motivating, not substituting for, the KNHANES analysis.

## 6.3 Clinical implications

The defensible reading is specific. In adults with *non-urgent, stable* tinnitus, perceived stress is a common co-travelling factor, which supports attention to stress, sleep, and mood in tinnitus counseling and cognitive-behavioral or sound-therapy approaches (Tunkel et al., 2014), consistent with stress tracking the tinnitus *symptom* more than the *threshold*. Equally important as a *negative* implication, the data do *not* license attributing acute, sudden, or asymmetric hearing loss to "stress": the association is with the symptom, not the cochlear threshold, and such presentations warrant prompt audiological and otolaryngological assessment regardless of psychosocial context. Mislabeling them as stress-related is precisely the hazard this work guards against—and the deterministic red-flag referral-safety layer that operationalizes that guard (so a low predicted risk can never suppress urgent referral for time-critical sudden sensorineural hearing loss) is a prospective component developed, with results, in the companion SAFE-EAR paper (Chandrasekhar et al., 2019).

## 6.4 Comparison with prior work and limitations

Unlike single-country prevalence studies (Park et al., 2014; Joo et al., 2015), STARS separates the tinnitus symptom from the audiometric threshold, prespecifies its analyses and sensitivity contrasts, and treats reproducibility as a first-class result. Key limitations bound the primary finding: (1) the design is *cross-sectional*, so it establishes association, not temporal order or causation, and cannot exclude reverse causation; (2) exposure and outcome are single self-report items (biasing associations toward the null); (3) the KNHANES examination provides air-conduction thresholds and tympanometry but not bone conduction, so residual conductive misclassification may remain and better-ear PTA understates unilateral/asymmetric disease (motivating the worse-ear and high-frequency analyses); (4) treating depressed mood and sleep as mediators rests on assumptions that may be violated; (5) the 40–69 band excludes the ≥70 group with the highest tinnitus/hearing burden, limiting generalizability; and (6) the cross-national transportability, AI-extraction, and referral-safety components are prospective here (design only; Supplementary Methods S1–S2) and are developed in the companion SAFE-EAR paper. These bound, but do not overturn, the prespecified primary finding.

# 7 Conclusion

STARS reports a survey-weighted, reproducible result: in representative Korean adults aged 40–69, perceived stress is associated with the tinnitus *symptom* but not, once occupational noise and age are accounted for, with the audiometric *threshold*. This symptom-versus-threshold dissociation was prespecified (H1), is reproducible end-to-end from committed code and data mappings, and is directionally echoed by NHANES under a related distress construct. Because the design is cross-sectional, it characterizes associations—not causes—and cannot resolve whether stress precedes tinnitus or bothersome tinnitus generates distress. Around this finding, STARS also fixes, in advance, a clinically grounded companion program: cross-national external validation with recalibration, common-support, and subgroup-fairness checks; and—developed with results in the companion SAFE-EAR paper—a clinician-verified, schema-constrained AI-extraction component and a deterministic red-flag safety layer, in which any prediction model is a research risk-stratification tool with declared intended use, never a diagnostic device. By releasing code, model cards, safety tests, and a synthetic-data path, the study maximizes transparency and reproducibility while avoiding unsupported causal claims. A prospective hospital-based clinical cohort remains necessary to study treatment timing and recovery—above all, to protect the narrow window in which sudden hearing loss is treatable.

# Declarations

**Author Contributions (CRediT):** *Geunsik Lim*—Conceptualization, Methodology, Software, Formal analysis, Data curation, Writing (original draft & editing), Visualization. The author read and approved this draft.

**Corresponding Author:** Geunsik Lim, Sungkyunkwan University, Republic of Korea. Email: `leemgs@g.skku.edu`.

**Funding:** No external funding was received for this study. Any funding for the prospective clinical extension will be disclosed at that stage.

**Ethics Approval:** Stages 1–2 use publicly available, fully deidentified survey microdata (KNHANES, NHANES) under each provider's data-use terms and do not constitute new human-subjects research. The Stage-3 clinical extension will obtain institutional review board approval at the participating clinical site before any data collection.

**Reporting Guidelines:** This empirical article follows STROBE (von Elm et al., 2007); the completed STROBE checklist (`paper/checklists/STROBE_checklist.md`) is provided as a supplementary file and submitted with the manuscript. The prospective prediction and referral-safety components follow TRIPOD+AI (Collins et al., 2024) and are developed, with their reporting checklist (`paper02/checklists/TRIPOD-AI_checklist.md`), in the companion SAFE-EAR paper; the clinical extension will follow SPIRIT.

**Analysis-Plan Status and Registration:** This is a survey-weighted research study whose prespecified primary KNHANES perceived-stress analyses are reported here

(Section 5), with NHANES as directionally consistent supporting evidence; the external-validation, AI-extraction, and clinical-extension components remain prospective (design only). The prespecified analysis plan (cycles, variables, adjustment sets, endpoints, and metrics) is time-stamped in the versioned repository.

**Data Availability:** KNHANES data are available from the Korea Disease Control and Prevention Agency subject to official access procedures; NHANES public-use files are available from the U.S. CDC. The repository provides code, configuration, variable-mapping templates, and a synthetic-data generator, but does not redistribute restricted raw data.

**Code Availability:** All analysis code, configuration, model cards, prompt templates, and the reporting checklist are openly available at https://github.com/leemgs/stars-audiology under an open-source license.

**AI-Use Disclosure:** Generative AI tools assisted in drafting and structuring this manuscript and the code scaffold; the author reviewed and is responsible for all content. In the proposed study, open medical language models will be used only for research feature extraction and guideline-linked explanation from deidentified text, always with clinician verification before analysis and never for autonomous clinical decisions.

**Conflicts of Interest:** The author declares no competing interests.

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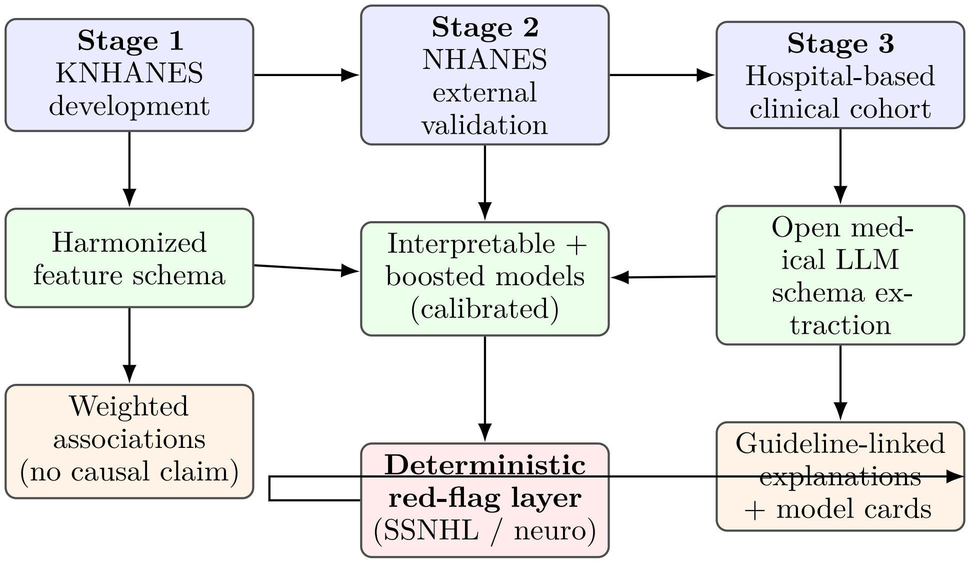


Figure 1: STARS conceptual framework. Public survey data (blue) feed a harmonized schema and calibrated, interpretable models (green); an open medical language model performs schema-constrained extraction in the clinical extension. A deterministic red-flag safety layer (red) can override probabilistic output to force urgent referral, so a low predicted risk never suppresses time-critical SSNHL care. Outputs (orange) are associations and guideline-linked explanations, not diagnoses.

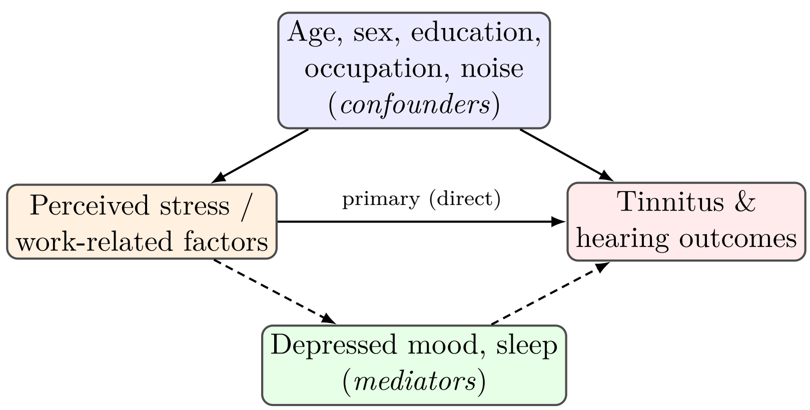


Figure 2: Assumed causal directed acyclic graph (DAG). The minimal sufficient adjustment set (blue confounders) is used in the *primary* analysis. Depressed mood and sleep (green) are treated as *mediators* of the stress→outcome path (dashed); conditioning on them (extended model) is a mediator-adjusted, potentially overadjusted estimate reported separately. Restricting to employed adults can open a selection/collider path, so all-adult analysis is primary and employed-restricted analysis is secondary.

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